

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

This observation project is to calculate the rotational period of a given asteroid and compare our findings with existing archival data. This observation is interesting as we get to both observe and calculate the rotation period and light curve amplitude to constrain the elongation of the asteroid, learning about the physical properties of small Solar System bodies. We will then verify our findings with previously calculated values, allowing us to gain experience with hands-on astronomical observation. All possible targets are in the main-belt with a magnitude from 13-15, and our main target will be 918 Itha. We will use the Mont4k CCD on the Kuiper 61'' telescope. They will be 60 second exposures with an I-band filter. With each image, we are comparing the flux to the same reference stars to find the relative brightness throughout the rotation. By measuring the changes in this light curve over time, we are able to uncover the science of rotational periods and asteroid elongation/shape.

Telescope Requested (0.5 points): Kuiper 61''

Instrument (0.5 points): Mont4K

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

With our target having a magnitude of ~ 14 , we will need the moon to be quite dark in order for us to measure the light curve. Ideally, it would be a new moon or close to a new moon. This would also eliminate the issue of changing brightness from the moon.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

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Object Name	Right Ascension	Declination	Apparent V magnitude	Object Type	Visibility
155 Scylla	00 51 54.10	-04 19 27.3	14.167	Asteroid	Sept - Nov
170 Maria	22 50 43.81	+13 26 04.4	13.209	Asteroid	Aug - Oct
188 Menippe	03 02 24.26	+24 23 46.2	13.391	Asteroid	Oct - Dec
167 Urda	01 36 58.47	+07 56 15.8	13.302	Asteroid	Sept - Nov
248 Lameia	02 33 32.76	+18 50 51.8	14.063	Asteroid	Oct - Dec
295 Theresia	01 39 24.80	+14 59 13.3	13.162	Asteroid	Sept - Nov
2 Pallas	01 19 37.37	-14 59 29.9	8.265	Asteroid	Sept - Nov
236 Honoria	23 05 13.51	-04 08 34.1	11.4	Asteroid	Aug - Oct
71 Niobe	23 12 31.07	+22 53 29.8	11.9	Asteroid	Aug - Oct
695 Bella	23 31 20.56	+24 18 41.0	12.0	Asteroid	Sept - Nov
106 Dione	23 41 02.37	-08 42 08.8	11.8	Asteroid	Sept - Nov
918 Itha	01 16 01.18	+25 32 34.5	14.0	Asteroid	Sept - Nov

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

We estimate an SNR of 150 in order to measure our target. Since we are measuring minute changes in an asteroid's brightness, we need very high precision. 918 Itha has a relatively short rotation period of around 3.5 hours. We plan to observe two full rotational periods, requiring around 7 hours of observation time.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

After one rotation, the asteroid should return to the same initial flux, creating a plottable curve that will repeat with each rotation. Averaging the results from the two rotational periods versus the initial flux will give us a final measurement of the rotational period that can be matched against existing data.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. (0.5 points)

If weather conditions do not permit us from making our observations, there is lots of published data on asteroids that we can pull from to use for our results.

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<https://sbn.psi.edu/pds/resource/lc.html>

<https://iopscience.iop.org/article/10.1088/1538-3873/ae7cb6/pdf>